Power Budget: Force Sensing Board

Team Number:	103
Project Name:	**GripRx**
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Version:	**1 (Zice Sun, 2026-10-16)**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano (MCU board)	DM164150	1.8 - 5.5 V	1	350.0	350.0	mA
	Instrumentation amplifier	INA125P	2.7 - 36 V	1	12.5	12.5	mA
	Quad op amp (low-pass filter)	MCP6004-I/P	1.8 - 6.0 V	1	30.0	30.0	mA
	Load cell (bridge excitation)	3135_0 (Phidgets)	5 V max	1	5.0	5.0	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design. Add additional power rails or change the power rail voltages if needed.

+12V Power Rail (not used)	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	None on this board	—	—	0	0	0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	0	mA
c1. Regulator or Source Choice	Not used (no +12 V loads)	—	—	0	0	0	mA
					Total Remaining Current Available on +12V Rail	0	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano (MCU board)	DM164150	1.8 - 5.5 V	1	350.0	350.0	mA
	Instrumentation amplifier	INA125P	2.7 - 36 V	1	12.5	12.5	mA
	Quad op amp (low-pass filter)	MCP6004-I/P	1.8 - 6.0 V	1	30.0	30.0	mA
	Load cell (bridge excitation)	3135_0 (Phidgets)	5 V max	1	5.0	5.0	mA
						0	mA
					Subtotal	397.5	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	496.9	mA
c2. Regulator or Source Choice	+5V linear regulator	L7805ABV-DG	7 - 35 V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	1003.1	mA

-5V Power Rail (not used)	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	None: all op amps are single-supply	—	—	0	0	0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on -5V Rail	0	mA
c3. Regulator or Source Choice	Not used (no negative rail)	—	—	0	0	0	mA
					Total Remaining Current Available on -5V Rail	0	mA

+3.3V Power Rail (not used)	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	None on this board	—	—	0	0	0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	0	mA
c4. Regulator or Source Choice	Not used (no +3.3 V loads)	—	—	0	0	0	mA
					Total Remaining Current Available on 3.3V Rail	0	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power sources, list each separately below and indicate which regulators will be connected to each supply. Confirm that the Total Remaining Current Available on each power source below is not negative.

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	9 V 3 A plug-in wall adapter (switching), issued in class	BestCH 9V 3A (Amazon B09ZTKTLGW)	100 - 240 VAC	+9V	3000	3000	mA
Power Rails Connected to External Power Source 1	+12V regulator (not used)	—	—	0	0	0	mA
	+5V linear regulator	L7805ABV-DG	7 - 35 V	1	1500	1500	mA
	+3.3V regulator (not used)	—	—	0	0	0	mA
		Total Remaining Current Available on External Power Source 1				1500.0	mA

External Power Source 2 (not used)	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	Not used (single 9 V adapter)	—	—	—	0	0	mA
Power Rails Connected to External Power Source 2	-5V regulator (not used)	—	—	0	0	0	mA
		Total Remaining Current Available on External Power Source 2				0	mA

E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.

	Component Name	Part Number	Supply Voltage Range	Capacity (mAh)	Required By Regulators	
	No battery (wall-powered board)	—	—	N/A	0	
				Battery Life	N/A	

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit

Absolute maximum currents are worst-case datasheet limits: PIC18F57Q43 maximum current on VDD pin 350 mA (48-pin device, -40 to +85 C); INA125P quiescent current 0.525 mA max plus 12 mA output short-circuit current; MCP6004 30 mA maximum current at supply pins; load cell 5 V excitation across its 1 kOhm bridge = 5 mA.

Worst-case L7805ABV-DG dissipation: (9 V - 5 V) x 497 mA = 2.0 W, so the regulator gets a Boyd 507302B00000G TO-220 heatsink. Expected normal draw is under 100 mA (about 0.4 W).